JOURNAL OF VERBAL LEARNING AND VERBAL BEHAVIOR 10, 176–181 (1971)

Recognition Memory for Sentence Meaning and Wording[1]

IAN BEGG[2]

Department of Psychology, University of Western Ontario, London 72, Ontario, Canada

Several hundred sentences were auditorily and visually presented to *S*s in a continuous recognition paradigm. The test sentences were either identical to the original or were paraphrases. Subjects were required to judge both the meaning and the wording of the test sentences as old or new. Meaning judgments decreased in accuracy as a function of lag, while the accuracy of wording judgments was uncorrelated to either the meaning judgments or lag. Meaning judgments were no less accurate if the test sentence was a paraphrase than if it was identical to the original, although the accuracy of wording judgments was above chance. Meaning judgments were affected relatively little by modality, while wording judgments were more accurate in visual presentations. Also, meaning and wording judgments were stochastically independent of each other.

Begg & Paivio (1969) hypothesized that the meaning of a concrete sentence is represented in memory as a nonverbal spatial image, while the individual words are represented as a sequential verbal string. They further speculated that "under natural conditions, *S*s rely exclusively on the imaginal representation of concrete sentences, thereby losing the individual words completely [p. 822]." If the *S* were later required to retrieve the sentence words, he would presumably do so by generating a verbal description of the retrieved image of the sentence meaning. The present experiment used naturally occurring, relatively concrete sentences in order to examine further implications of the model with respect to memory for sentence meaning and wording.

The idea that the wording of a message is reconstructed from the meaning is not new. It was central to the theoretical accounts of Bühler (1908) and Bartlett (1932), while the idea that a meaning or theme could be represented as an image was suggested as a possibility by Pompi and Lachman (1967), and was tested and extended by Yuille and Paivio (1969), who found that the concreteness and thematicity of a passage interacted in determining the number of words from the passage subsequently recalled. That is, more words were recalled from thematic than nonthematic passages, but only when the passages were concrete. Similarly, the idea that the details of sentence wording are not specifically represented in memory after comprehension has occurred, was suggested by Sachs (1967a, b) and Bregman and Strasberg (1968), although they were not concerned with imaginal or retrieval processes.

There are other implications of the general theoretical framework above that have not been specifically tested, although some data exist. For example, memory for wording should be independent of the passage of either time or interpolated activity, except to the extent that the image itself might not be retrieved at longer delays. Sachs (1967a,b) found that the probability of a change in

[1] The research was conducted while the author held a postgraduate scholarship from the National Research Council of Canada, at the University of Western Ontario. The author would like to acknowledge the assistance given by W. A. Wickelgren, in whose laboratory the experiment was designed, and A. Paivio, in whose laboratory the data were collected and who offered invaluable critical comments during the preparation of the manuscript.

[2] Requests for reprints should be sent to Ian Begg, Department of Psychology, University of Western Ontario, London 72, Ontario, Canada.

meaning being reported decreased from about .90 to .80 over a range from about 0–50 sec, while the report of wording changes dropped from an immediate level of about .90 to an asymptotic level of .60 after about 7.5 sec. The results could be interpreted to mean that the *wording information is stored for a very short time, possibly long enough for comprehension to take place, after which wording memory is a process of reconstruction.* A further implication is that memory for wording is independent of either the passage of time or memory for meaning, beyond 7.5 sec.

In the present study, the range of lags was increased to include from 2–64 intervening sentences between the original presentation of a sentence and the test presentation of the sentence or a paraphrase for meaning or wording judgments. In addition, both visual and auditory presentations were used. Several predictions are clear from the interpretation or model given in the preceding paragraph. The accuracy of meaning judgments should decrease systematically as a function of lag, and should thus be highly negatively correlated with the number of intervening items. However, the accuracy of wording judgments should be uncorrelated with either the meaning judgments or lag. Additionally, the accuracy of meaning and wording judgments should be stochastically independent, that is, the product of the probabilities of correct meaning and wording judgments should approximate the probability of a jointly correct judgment of meaning and wording, since meaning judgments depend on the retrievability from memory of an image of the sentence meaning, and wording judgments depend on the match between the generated description of the image and the wording of the test sentence. Furthermore, meaning judgments should not be less accurate if the test sentence is a paraphrase than if it is identical to the original. Although the paraphrase lacks exact syntactic and lexical cues for retrieving the original sentence meaning, such cues are irrelevant in the present model, in which the appropriate cue is presumed to be the image that represents the meaning of the test sentence.

The modality variable was included primarily because of anecdotal data suggesting that grammar may be a more salient feature of writing than of speech. In the case of speech, gestures and inflections assume much of the burden of emphasizing different aspects of the meaning of a message, while in writing the emphasis is provided by syntactic and lexical features of sentence structure. Additionally, even competent writers do not necessarily speak grammatically (cf. Maclay & Osgood, 1959). Sachs (1967a) suggested, and Wanner (1968) demonstrated, that increasing the emphasis on sentence wording produces incremental accuracy in wording judgments, while Bregman and Strasberg (1968) suggested that nonsemantic information might cause changes in the meaning trace, and so would be stored implicitly in situations where such effects might occur. These considerations led to the prediction that, if grammatical features are indeed more salient in writing than in speech, wording judgments should be more accurate for visually presented than auditorily presented sentences, despite Murdock's (1967) evidence that auditory presentations of verbal material result in superior memory performance than visual presentations. The accuracy of meaning judgments should not be affected by modality, since the central feature of any message is the meaning.

In the present study, then, several hundred sentences were presented auditorily or visually to *S*s. The sentences were presented again with the same or altered wording, after a number of intervening sentences, at which time the *S*s were required to judge whether the meaning or wording of the test presentation was new or old.

METHOD

Subjects

Sixteen female volunteers from the University of Western Ontario served as *S*s. Eight served in the memory experiment, and eight rated the sentences on imagery.

Materials

Two parallel lists of 502 sentences, A and B, were constructed. Since A and B differed only in the exact sentences used, only the construction of one list will be described. Two hundred sixty sentences were selected from an existing pool of 3600 sentences, which were originally selected randomly from novels, with the only constraints being that they were relatively concrete and contained no direct discourse or names of people. The sentences varied in length from 35–80 characters, and represented a variety of syntactic types, presumably in the proportion with which those types occur in writing. Fifty of the sentences were chosen to be practice sentences, 18 to be fillers, and 192 to be test sentences. Each sentence was rewritten with minimal alteration of the original wording. The wording changes represented the formal (order, addition, deletion, and voice change) and lexical (synonym substitution) changes reported by Sachs (1967b). For example, *After talking for a moment, the boy hung up the phone* was rewritten as *After talking for a moment, the boy hung the phone* up. The 192 test sentence pairs, each pair consisting of the originally selected sentence and its paraphrased counterpart, were separated into four groups. In two groups, the original sentence was chosen as the New sentence, and in two the rewritten version was chosen to be New. One group of each of the above two was chosen for the Identical (Id) condition, while the remaining two were chosen for the Paraphrase (Para) condition. A test sequence consisted of 402 sentences, and was constructed using the 18 fillers, the 192 New sentences, 96 Id sentences (identical to 96 of the New sentences), and 96 Para sentences (paraphrases of the remaining 96 New sentences). Each Id or Para sentence occurred later in the list than its New counterpart, by either 2, 4, 8, 16, 32, or 64 intervening sentences, with 16 pairs in each delay condition, except for the 4- and 8-item lags in the Para condition, where there were 15 and 17 respectively. The 100 sentences in the practice sequence were similarly arranged with 50 new, 25 Id, and 25 Para sentences.

Lists A and B were tape recorded with one sentence occurring every 10 sec. The average recording time for a sentence was about 5 sec, and a click preceded each sentence by about 1 sec, so that intervals of about 4 sec separated the successive sentences on the tape. The sentences were also typed with 25 to a page. No sentence was longer than one typed line.

Procedure

Memory Experiment. Each *S* heard one list of sentences and silently read the other. Half the *S*s heard List A while half heard List B. The order of presentation of the visual and auditory material was counterbalanced.

The *S*s were instructed to listen to or read the sentences silently as they occurred. After each sentence, the *S*s made two judgments. The first judgment was a "Yes–No" response depending on whether or not the *S* recognized the meaning of the sentence as one with which he had been previously presented. Disregarding the 18 fillers, 192 of the 384 sentences, the New sentences, thus required a "No" response, while the Id and Para sentences required a "Yes" response. After the meaning judgment, the *S* made a second "Yes–No" judgment, this time depending on whether he recognized the exact words in the sentence as being identical to the words in an earlier sentence. If the sentence was New, the correct response was "No"–"No" (N). For the Id sentences, the correct response was "Yes"–"Yes" (I), while for the Para sentences, "Yes"–"No" (P) was correct.

The presentation rates for the visual and auditory material were 5- and 10-sec per sentence, respectively. The rates differed since it takes longer to listen to a sentence than to read it. Pilot work indicated that an average of 1- to 2-sec reading time was necessary for sentences comparable to the present ones. Thus, about 3- to 4-sec response time was allowed between successive sentences in each modality. In the visual condition, the *S*s slid a cardboard rider down the typed pages, paced by a signal from the experimenter. A hole in the rider was large enough to permit the viewing of one sentence at a time. The *S*s were tested individually, and made their responses orally. They were given a 15-min coffee break between Lists. The entire session lasted about 3.5 hr.

Ratings. Each typed List was presented to four *S*s to rate the image-arousing capacity (*I*) of the sentences on a 7-point scale, from low to high. The instructions were modified from Paivio, Yuille, and Madigan (1968).

Results

Wording Judgments

The I responses ("Yes"–"Yes") were analyzed by a $2 \times 6 \times 2$ analysis of variance with Order (List A visually, B auditorily, or the reverse) as the independent factor, and Lag (2, 4, 8, 16, 32, or 64 intervening items) and Modality (auditory or visual) as repeated factors. All factors except *S*s had fixed levels. The dependent factor was P(I|Id) − P(I|Para), or hits minus false alarms. In all 24 cells P(I|Id) was greater than P(I|Para), so that I responses were above chance accuracy, $z = 4.90$, $p < .01$.

The only significant F ratios were the main effects of Lag, $F(5, 30) = 3.44$, $p < .025$, and Modality, $F(1, 6) = 11.3$, $p < .025$. The accuracy scores, as a function of Lag from shortest to longest, were .323, .218, .385, .193, .329, and .262, indicating that the Lag effect was not systematic. The product-moment correlation between the above scores and the number of intervening items was −.107, which does not differ significantly from zero. The Modality effect indicated that accuracy was, as predicted, superior with visual presentations (.331 compared to .236).

The P responses ("Yes"–"No") were analyzed by a 2 × 6 × 2 analysis of variance with the same factors as above. The dependent factor was P(P|Para) − P(P|Id). The accuracy was again above chance, with hits greater than false alarms in all 24 cells, $z = 4.90$, $p < .01$.

The results of the analysis of variance were also as above, with the main effect of Lag, $F(5, 30) = 3.12$, $p < .025$, not systematic (.322, .225, .386, .216, .283, and .284). Accuracy and Lag were not significantly correlated ($r = -.096$), but the accuracy of P responses was highly related to the accuracy of I responses ($r = .934$, $p < .01$). The Modality effect, $F(1, 6) = 9.78$, $p < .025$, indicated that visual presentations were superior (.337) to auditory (.238).

The absolute levels of accuracy of report for wording, uncorrected for guessing, averaged across Modality and Id and Para sentences, $Pc(W) = \frac{1}{2}P(I|Id) + \frac{1}{2}P(P|Para)$, were calculated to allow comparison with Sachs' (1967a, 1967b) results. From shortest to longest lags, the proportions correct were .658, .605, .681, .579, .621, and .579, with an overall average of about .62, which is quite similar to Sachs' finding of .60.

Meaning Judgments

Meaning judgments, P(I or P|Id or Para) − P(I or P|New), were analyzed by a 2 × 6 × 2 × 2 analysis of variance with Order, Lag, and Modality as above, and Test (Id or Para) as the factors. A main effect of Lag, $F(5, 30) = 19.6$, $p < .01$, reflected monotonically decreasing accuracies as a function of increasing Lag. The scores, .965, .964, .952, .935, .917, and .858, correlated −.996, $p < .01$, with the number of intervening items, in contrast to the non-significant correlations between I and P responses and Lag. Neither I nor P responses were significantly correlated with meaning accuracy, $r = .126$ and .095, respectively. In the analysis of variance, the Order × Lag interaction, $F(5, 30) = 5.14$, $p < .01$, and the four-way interaction, $F(5, 30) = 4.17$, $p < .01$, were also significant. In 47 of the 48 conditions, accuracy was equal to or greater than .823, but, presumably because of chance fluctuation, when List B was presented visually, *S*s only identified .696 of the old meanings for Id sentences after 64 intervening items. Both interactions can be attributed to this one highly deviant point. The only qualification of the main effect of Lag is that the accuracy at the 64-item delay is underestimated.

Since there is a real possibility of a ceiling effect, the proportion correct for Id and Para test sentences, and for auditory and visual presentations were calculated. The means were only calculated across the five shortest Lags, since the interactions above indicate that the accuracy for Id sentences and visual presentations would be underestimated in the 64-item delay condition. Meaning judgments were not less accurate in Para than Id sentences (.947 vs. .937), or in auditory than visual presentations (.955 vs. .927). In fact, both differences were slightly in the opposite direction. The ceiling effect criticism, which implies that the failure to find more accurate meaning judgments in Id than Para, and visual than auditory presentations could be attributed to the restriction of the possible range of variability, is clearly not valid because of the above reversal in the direction of means.

Stochastic Independence

The proportions of correct meaning judgments, Pc(M) × 3072 = (N|New) + (I or P|Id

or Para), wording judgments, Pc(W) × 3072 = (N or P|New or Para) + (I|Id), and jointly correct judgments, Pc(M and W) × 3072 = (N|New) + (I|Id) + (P|Para), were calculated across all other variables. The respective proportions correct for Pc(M), Pc(W), and Pc(M and W) were .952, .817, and .790, while the value of the product Pc(M)Pc(W) was .778. Since the latter two values are approximately equal, the independence hypothesis is supported. Similarly, conditionalizing on the meaning being old, Pc(M) × 1536 = (I or P|Id or Para), Pc(W) × 1536 = (N or P|Para) + (I|Id), and Pc(M and W) × 1536 = (I|Id) + (P|Para). The respective numerical values were .932, .642, and .608, while Pc(M)Pc(W) = .598. Again the latter two values are nearly equal, supporting the independence hypothesis.

Ratings

The sentences were collapsed across Lists, so that between 30 and 34 sentences were rated in each Lag. The mean ratings for each Lag ranged from 4.50 to 5.08, for an overall average of 4.79 on a 7-point scale. The reliabilities between first and second presentations of the sentences ranged from .65 to .92, with an average of .81 (by a z transform) in the Id sentences, and with a range of .68 to .84, and an average of .77 in the Para sentences. The values .81 and .77 do not differ significantly, suggesting that imagery ratings reflect sentence meaning and not wording.

Discussion

Several facts are clear from the results of the experiment. First, although the accuracy of the wording judgments was consistently above chance, meaning judgments were not more accurate for identical test presentations than for paraphrases. Therefore, as predicted by the present model, syntactic and lexical cues did not provide any facilitation in the retrieval of the stored representation of sentence meaning. Secondly, the accuracy of meaning judgments was systematically related to the number of intervening items between presentation and test, while the accuracy of wording judgments was unrelated to the number of intervening items, and was independent of the accuracy of meaning judgments, both stochastically and correlationally. Additionally, the level of accuracy of wording judgments was similar to the results of Sachs (1967a,b), while the meaning judgments approximated the results of Shepard (1967). Thus, while *S*s have a remarkable capacity for remembering sentence meaning, at least when the sentences in the sample differ from each other in meaning, their memory for wording is not impressive. Finally, visual presentations were more conducive to accurate wording judgments than were auditory presentations, suggesting that grammar is a more salient feature of printed language than of speech.

The results generally support a model of sentence memory which hypothesizes that the meaning of a concrete sentence is represented in memory as a nonverbal spatial image, while the words are reconstructed from the image at retrieval. It should again be emphasized here, in agreement with the suggestions of Sachs (1967a,b) and Bregman and Strasberg (1968), that *S*s *can* store or encode the wording of a sentence explicitly, although they do not normally do so. The tenability of the imagery aspect of the model comes first from the use of predominantly concrete sentences in the experiment, and secondly from discussions with the memory *S*s. Each *S*, when asked how he managed to remember such a large number of sentences, replied that he remembered them as pictures or images.

At any rate, the conclusion from the present study is that memory for the meaning of concrete sentences is independent of memory for their wording. Any successful theory of memory for sentences must account for both types of memory, either by positing the existence of some kind of memory trace that remains constant or varies randomly from 7.5 sec to 11 min, or by assuming that sentence

words are normally reconstructed from meaning, and are not necessarily represented explicitly in memory.

References

Bartlett, F. C. *Remembering: A study in experimental and social psychology.* Cambridge, England: Cambridge University Press, 1932.

Begg, I., & Paivio, A. Concreteness and imagery in sentence meaning. *Journal of Verbal Learning and Verbal Behavior*, 1969, **8**, 821–827.

Bregman, A., & Strasberg, R. Memory for syntactic form of sentences. *Journal of Verbal Learning and Verbal Behavior*, 1968, **7**, 396–403.

Bühler, K. Tatsachen und Probleme zu einer Psychologie der Denkvorgange. *Arch. ges. Psychol.*, 1908, **12**, 1–92. Cited by Sachs (1967b).

Maclay, H., & Osgood, C. E. Hesitation phenomena in spontaneous English speech. *Word*, 1959, **15**, 19–44.

Murdock, B. B., Jr. Auditory and visual stores in short-term memory. *Acta Psychologica*, 1967, **27**, 316–324.

Paivio, A., Yuille, J. C., & Madigan, S. A. Concreteness, imagery, and meaningfulness values for 925 nouns. *Journal of Experimental Psychology*, 1968, **76**, Monograph Supplement No. 1, Part 2.

Pompi, K. F., & Lachman, R. Surrogate processes in the short-term retention of connected discourse. *Journal of Experimental Psychology*, 1967, **75**, 143–150.

Sachs, J. S. Recognition memory for syntactic and semantic aspects of connected discourse. *Perception and Psychophysics*, 1967, **2**, 437–442 (a).

Sachs, J. S. Recognition of semantic, syntactic, and lexical changes in sentences. Paper presented at Psychonomic Society, October, 1967 (b).

Shepard, R. N. Recognition memory for words, sentences, and pictures. *Journal of Verbal Learning and Verbal Behavior*, 1967, **6**, 156–163.

Wanner, H. E. On remembering, forgetting and understanding sentences: A study of the deep structure hypothesis. Unpublished doctoral dissertation, Harvard University, 1968.

Yuille, J. C., & Paivio, A. Abstractness and the recall of connected discourse. *Journal of Experimental Psychology*, 1969, **82**, 467–471.

(Received September 21, 1970)